

Chapter 1: Introduction

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Computing Environments - Traditional

- Stand-alone general purpose machines
- But blurred as most systems interconnect with others (i.e., the Internet)
- **Portals** provide web access to internal systems
- **Network computers (thin clients)** are like Web terminals
- Mobile computers interconnect via **wireless networks**
- Networking becoming ubiquitous – even home systems use **firewalls** to protect home computers from Internet attacks





Computing Environments - Mobile

- Handheld smartphones, tablets, etc
- What is the functional difference between them and a “traditional” laptop?
- Extra feature – more OS features (GPS, gyroscope)
- Allows new types of apps like *augmented reality*
- Use IEEE 802.11 wireless, or cellular data networks for connectivity
- Leaders are **Apple iOS** and **Google Android**





Computing Environments – Distributed

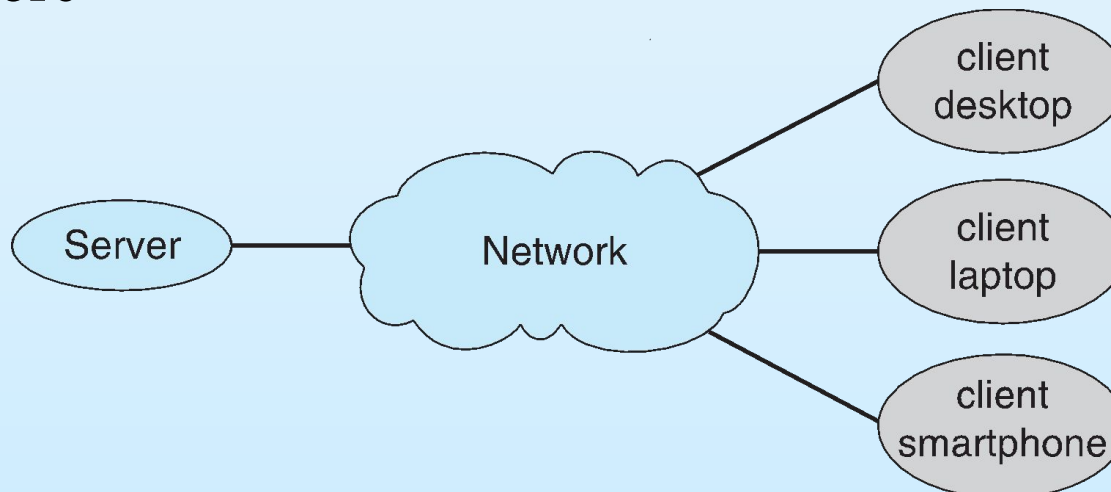
- Distributed computing
 - Collection of separate, possibly heterogeneous, systems networked together
 - 4 **Network** is a communications path, **TCP/IP** most common
 - **Local Area Network (LAN)**
 - **Wide Area Network (WAN)**
 - **Metropolitan Area Network (MAN)**
 - **Personal Area Network (PAN)**
 - **Network Operating System** provides features between systems across network
 - 4 Communication scheme allows systems to exchange messages
 - 4 Illusion of a single system





Computing Environments – Client-Server

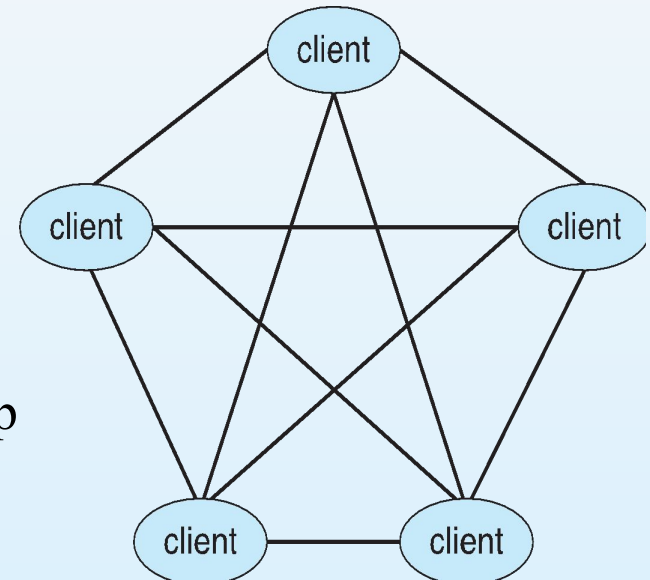
- Client-Server Computing
 - Dumb terminals supplanted by smart PCs
 - Many systems now **servers**, responding to requests generated by **clients**
 - 4 **Compute-server system** provides an interface to client to request services (i.e., database)
 - 4 **File-server system** provides interface for clients to store





Computing Environments - Peer-to-Peer

- Another model of distributed system
- P2P does not distinguish clients and servers
 - Instead all nodes are considered peers
 - May each act as client, server or both
 - Node must join P2P network
 - 4 Registers its service with central lookup service on network, or
 - 4 Broadcast request for service and respond to requests for service via *discovery protocol*
- Examples include Napster and Gnutella, **Voice over IP (VoIP)** such as Skype





Computing Environments - Virtualization

- Allows operating systems to run applications within other OSes
 - Vast and growing industry
- **Emulation** used when source CPU type different from target type (i.e. PowerPC to Intel x86)
 - Generally slowest method
 - When computer language not compiled to native code – **Interpretation**
- **Virtualization** – OS natively compiled for CPU, running **guest** OSes also natively compiled
 - Consider VMware running WinXP guests, each running applications, all on native WinXP **host** OS
 - **VMM** (virtual machine Manager) provides virtualization services





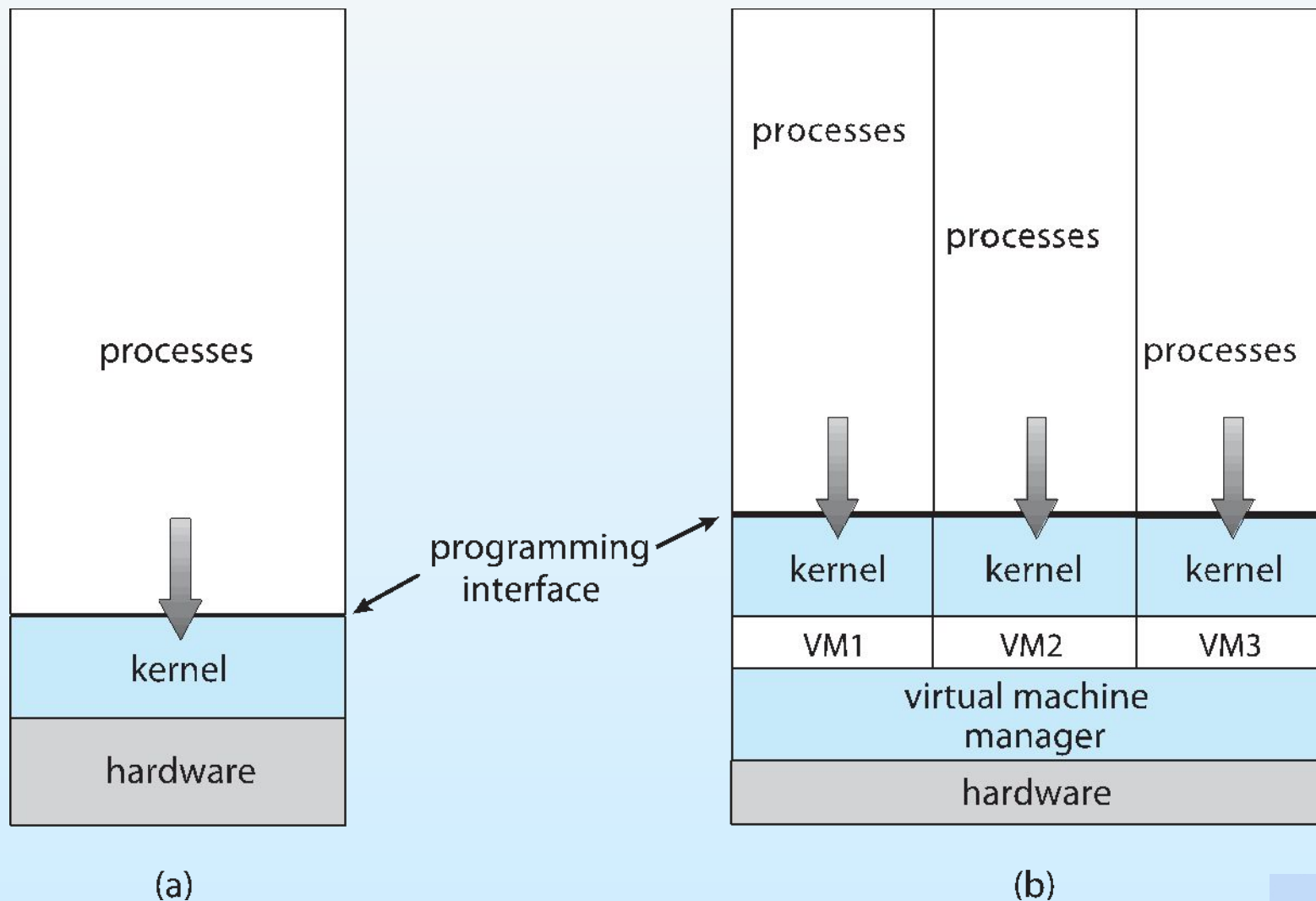
Computing Environments - Virtualization

- Use cases involve laptops and desktops running multiple OSES for exploration or compatibility
 - Apple laptop running Mac OS X host, Windows as a guest
 - Developing apps for multiple OSES without having multiple systems
 - QA testing applications without having multiple systems
 - Executing and managing compute environments within data centers
- VMM can run natively, in which case they are also the host
 - There is no general purpose host then (VMware ESX and Citrix XenServer)





Computing Environments - Virtualization





Computing Environments – Cloud Computing

- Delivers computing, storage, even apps as a service across a network
- Logical extension of virtualization because it uses virtualization as the base for its functionality.
 - Amazon **EC2** has thousands of servers, millions of virtual machines, petabytes of storage available across the Internet, pay based on usage
- **Many types:**
 - **Public cloud** – available via Internet to anyone willing to pay
 - **Private cloud** – run by a company for the company's own use
 - **Hybrid cloud** – includes both public and private cloud components





Computing Environments – Cloud Computing

- **Services:**

- **Software as a Service (SaaS)** – one or more applications available via the Internet (i.e., word processor)
- **Platform as a Service (PaaS)** – software stack ready for application use via the Internet (i.e., a database server)
- **Infrastructure as a Service (IaaS)** – servers or storage available over Internet (i.e., storage available for backup use)





Computing Environments – Real-Time Embedded Systems

- Real-time embedded systems most prevalent form of computers
 - Vary considerable, special purpose, limited purpose OS, **real-time OS**
 - Use expanding
- Many other special computing environments as well
 - Some have OSES, some perform tasks without an OS
- Real-time OS has well-defined fixed time constraints
 - Processing ***must*** be done within constraint
 - Correct operation only if constraints met





Open-Source Operating Systems

- Operating systems made available in source-code format rather than just binary **closed-source**
- Counter to the **copy protection** and **Digital Rights Management (DRM)** movement
- Started by **Free Software Foundation (FSF)**, which has “copyleft” **GNU Public License (GPL)**
- Examples include **GNU/Linux** and **BSD UNIX** (including core of **Mac OS X**), and many more
- Can use VMM like VMware Player (Free on Windows), Virtualbox (open source and free on many platforms - <http://www.virtualbox.com>)
 - Use to run guest operating systems for exploration



End of Chapter 1

